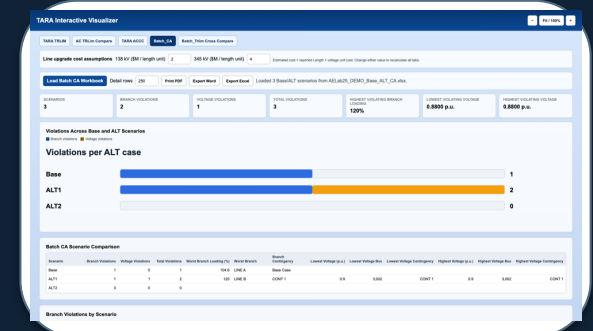
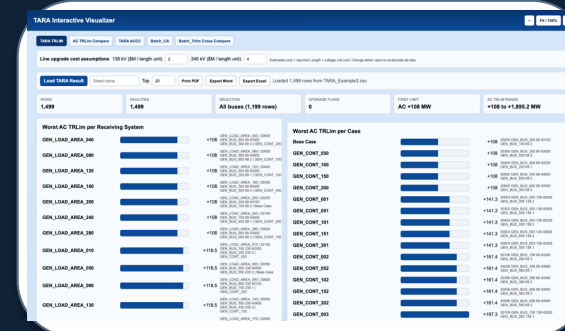
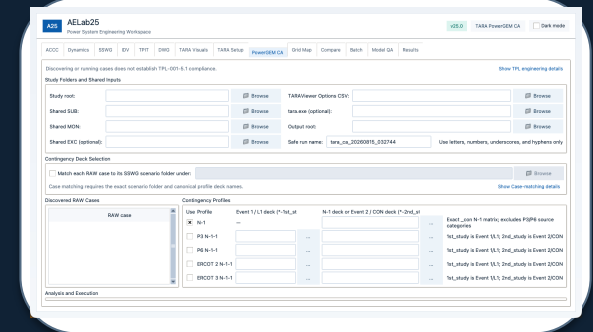
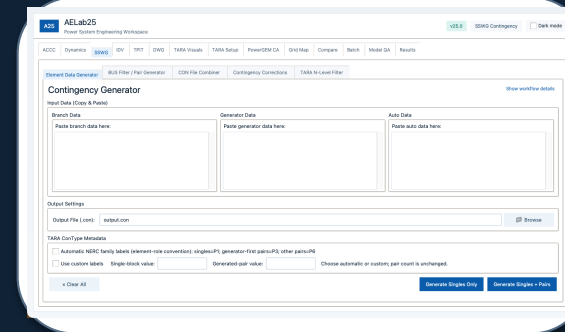


AELab25

Modern UI Overview

Power System Engineering Workspace

14 main workspaces
TARA data visualization • Base/ALT comparison



AELab25 • v25.0 • August 2026

Screens use blank or synthetic demonstration inputs.
They are not engineering study results.

Configure case inputs, criteria, execution, and progress tracking.

A25

AELab25

Power System Engineering Workspace

v25.0

ACCC Analysis

☐ Dark mode

ACCC

Dynamics

SSWG

IDV

TPIT

DWG

TARA Visuals

TARA Setup

PowerGEM CA

Grid Map

Compare

Batch

Model QA

Results

ACCC Input Files

Case File (.sav):

Browse

Subsystem File (.sub):

Browse

Monitor File (.mon):

Browse

Contingency File (.con):

Browse

ACCC Parameters

Thermal Loading Threshold (%):

90.0

Voltage High Limit (p.u.):

1.05

Voltage Low Limit (p.u.):

0.92

Voltage Deviation Threshold (p.u.):

0.07

Tolerance:

0.5

Number of Threads:

8

ACCC Method:

DSP

DSP: Stable (Recommended) | Parallel: Fast but unstable | Multiprocessing: Best for multiple cases

Number of Processes:

4

Output Excel File:

contingency_results.xlsx

Browse

ACCC Control

▶ Run ACCC Analysis

■ Kill Hanging PSSE

⬇ Export ACCC Results

■ PSSE Violations Report

Export: All

ACCC Progress

Prepare fault or event simulations and review dynamic-study outputs.

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Power System Engineering Workspace

v25.0

Dynamic Analysis

☐ Dark mode

ACCC

Dynamics

SSWG

IDV

TPIT

DWG

TARA Visuals

TARA Setup

PowerGEM CA

Grid Map

Compare

Batch

Model QA

Results

Fault Simulation

Dynamic Simulator

Dynamic Input Files

Case File (.sav):

 Browse

Dynamics File (.dyr):

 Browse

Converted Case File (.sav):

 Browse

Snapshot File (.snp):

 Browse

Dynamic Simulation Parameters

Fault Bus Number:

Fault Start Time (s):

Fault Duration (s):

Total Simulation Time (s):

Fault Impedance (p.u.):

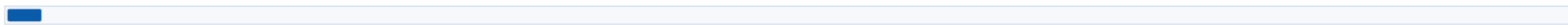
Output File (.outx):

 Browse☐ Enable Auto-Plotting

Dynamic Control

 **Run Dynamic Analysis** Stop Analysis View Dynamic Results

Dynamic Progress



Generate, combine, correct, and N-level-filter contingency decks.

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Power System Engineering Workspace

v25.0

SSWG Contingency

☐ Dark mode

ACCC

Dynamics

SSWG

IDV

TPIT

DWG

TARA Visuals

TARA Setup

PowerGEM CA

Grid Map

Compare

Batch

Model QA

Results

Element Data Generator

BUS Filter / Pair Generator

CON File Combiner

Contingency Corrections

TARA N-Level Filter

Contingency Generator

[Show workflow details](#)

Input Data (Copy & Paste)

Branch Data

Paste branch data here:

Generator Data

Paste generator data here:

Auto Data

Paste auto data here:

Output Settings

Output File (.con): output.con

 Browse

TARA ConType Metadata

☐ Automatic NERC family labels (element-role convention): singles=P1; generator-first pairs=P3; other pairs=P6☐ Use custom labels Single-block value:

Generated-pair value:

Choose automatic or custom; pair count is unchanged.

 Clear All

Generate Singles Only

Generate Singles + Pairs

Build structured PSS/E IDV actions from equipment inputs.

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Power System Engineering Workspace

v25.0

IDV Generator

☐ Dark mode

ACCC

Dynamics

SSWG

IDV

TPIT

DWG

TARA Visuals

TARA Setup

PowerGEM CA

Grid Map

Compare

Batch

Model QA

Results

CSV Files

 Browse - Line Impedance Estimator CSV

 Browse - Ratings CSV (Optional)

Enter Data

From Bus:

To Bus:

Min Rating (Amps):

Length (Miles):

Circuit ID:

TL Type:

TL Upgrade

Voltage Level (kV):

Conductors Per Phase:

Geometry:

Shield:

Environmental Index:

Construction Type:

Conductor Type:

1

New Construction

▶ Generate IDV

↕ Export IDV

Export Settings

IDV Filename:

.idv

Generated IDV

Bring planning datasets together and monitor processing output.

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AELab25

Power System Engineering Workspace

v25.0TPIT☐ Dark mode

ACCCDynamicsSSWgidvTPITDWGTARA VisualsTARA SetupPowerGEM CAGrid MapCompareBatchModel QAResults

TPIT - Transmission Project Information Tool

Input Files

Planning Data Export:

Browse

Existing Report:

Browse

MOD Report:

Browse

Project Completion Report:

Browse

▶ Process Files

Processing Output

Configure and run DWG contingency workflows from one workspace.

A25

AELab25

Power System Engineering Workspace

v25.0

DWG Contingency

☐ Dark mode

ACCC

Dynamics

SSWG

IDV

TPIT

DWG

TARA Visuals

TARA Setup

PowerGEM CA

Grid Map

Compare

Batch

Model QA

Results

DWG Contingency Tool

Run con2xls, dwg_ctg, and Oncor P5 from one window.

[↻ Refresh](#)**Setup**

con2xls Builder

DWG CTG Builder

Workflow

Run Log

Tool folder

 Browse[↗ Open](#)

Python command

 Browse[↗ Open](#)

Working folder

 Browse[↗ Open](#)☒ Run each step from its script folder

con2xls INI

 Browse[↗ Open](#)

DWG CTG INI

 Browse[↗ Open](#)

Oncor P5 INI

 Browse[↗ Open](#)

CON file

 Browse[↗ Open](#)

Excel file

 Browse[↗ Open](#)

Output folder

 Browse[↗ Open](#)Common extra arguments

Discovery complete.

Start the local interactive visualizer for TARA result files.

A25

AELab25

Power System Engineering Workspace

v25.0

TARA Visualizer

☐ Dark mode

ACCC

Dynamics

SSWG

IDV

TPIT

DWG

TARA Visuals

TARA Setup

PowerGEM CA

Grid Map

Compare

Batch

Model QA

Results

Interactive Browser

TARA TRLIM

TARA ACCC

Batch_CA

Batch_Trlim Cross Compare

Interactive Server

Host

127.0.0.1

Port

8765

▶ Start Server

■ Stop Server

➤ Open Browser

Status Stopped

URL

Runs the TARA browser visualizer from AELab25. The browser page includes TARA TRLIM and TARA ACCC views for CSV, TXT, XLSB, XLSX, or XLSM inputs, plus single-workbook Batch_CA and Batch_Trlim Cross Compare views for XLSX or XLSM inputs.

Compare cases and prepare INCH and model-change packages.

A25

AELab25

Power System Engineering Workspace

v25.0

TARA Case Preparation

☐ Dark mode

ACCC

Dynamics

SSWG

IDV

TPIT

DWG

TARA Visuals

TARA Setup

PowerGEM CA

Grid Map

Compare

Batch

Model QA

Results

TARA / PyPowerGEM runs only where its licensed Windows runtime is available. Source cases are read-only; outputs use new paths unless Overwrite is selected.

Compare solve checks are informational; use Case Diagnostics for warning and error review.

[Show Diagnostics policy details](#)

Compare Cases → INCH

Apply INCH → New Case

Build INCH from Inputs

Case Diagnostics

Case Pair (RAW/EPC)

Source case:



Browse

Version:

35

Target case:



Browse

Version:

35

Output folder:



Browse

TARA DLL (optional):



Browse

Output prefix:

transform

Both source→target and target→source INCH files are generated.

Outputs and Validation



Check whether both cases solve (informational only)



Export comparison reports



Apply and recompare forward transform



Overwrite existing outputs



Generate INCH Files

Ready.

Difference Summary

Equipment

Modified

Source only

Target only

Discover cases, match decks, preview jobs, run CA, and consolidate results.

A25

AELab25

Power System Engineering Workspace

v25.0TARA PowerGEM CADark mode

ACCCDynamicsSSWVIDVTPITDWGTARA VisualsTARA SetupPowerGEM CAGrid MapCompareBatchModel QAResults

Discovering or running cases does not establish TPL-001-5.1 compliance.

Show TPL engineering details

Study Folders and Shared Inputs

Study root:

Browse

TARAViewer Options CSV:

Browse

Shared SUB:

Browse

tara.exe (optional):

Browse

Shared MON:

Browse

Output root:

Browse

Shared EXC (optional):

Browse

Safe run name:

tara_ca_20260815_032744

Use letters, numbers, underscores, and hyphens only

Contingency Deck Selection

☐ Match each RAW case to its SSWG scenario folder under:

Browse

Case matching requires the exact scenario folder and canonical profile deck names.

Show Case-matching details

Discovered RAW Cases

RAW case

Contingency Profiles

Use Profile	Event 1 / L1 deck (*-1st_st		N-1 deck or Event 2 / CON deck (*-2nd_sl		
☒ N-1	—			...	Exact _con N-1 matrix; excludes P3/P6 source categories
☐ P3 N-1-1		...		...	1st_study is Event 1/L1; 2nd_study is Event 2/CON
☐ P6 N-1-1		...		...	1st_study is Event 1/L1; 2nd_study is Event 2/CON
☐ ERCOT 2 N-1-1		...		...	1st_study is Event 1/L1; 2nd_study is Event 2/CON
☐ ERCOT 3 N-1-1		...		...	1st_study is Event 1/L1; 2nd_study is Event 2/CON

Analysis and Execution

Launch the geographic planning workspace for network context.

A25

AELab25

Power System Engineering Workspace

v25.0

Geographic Map

☐ Dark mode

ACCC

Dynamics

SSWG

IDV

TPIT

DWG

TARA Visuals

TARA Setup

PowerGEM CA

Grid Map

Compare

Batch

Model QA

Results

Geographic Grid Map & Planning

Open the standalone geographic workspace for public transmission lines, substations, power plants, PSS/E generators and loads, shunts, transformers, bus coordinates, and proposed equipment drawing.

▶ Start & Open Geographic Map

↗ Open Geographic Map

Stopped

Compare scenario violations under consistent thermal and voltage criteria.

A25

AELab25

Power System Engineering Workspace

v25.0

ACCC Comparison

☐ Dark mode

ACCC

Dynamics

SSWG

IDV

TPIT

DWG

TARA Visuals

TARA Setup

PowerGEM CA

Grid Map

Compare

Batch

Model QA

Results

ACCC result file or folder

File

Folder

Output folder

Browse

Dashboard title

ACCC Results Comparison

Thermal flag (%)

100.0

Low voltage flag

0.9

Reads contingency_results.xlsx or the matching thermal, voltage, voltageDeviations, and NotConverged CSV exports, then counts violations per case.

▶ Generate Comparison

↗ Open Dashboard

Open Output Folder

Dashboard

Apply scripts across multiple cases with traceable status.

A25

AELab25

Power System Engineering Workspace

v25.0

Batch Case Scripts

☐ Dark mode

ACCC

Dynamics

SSWG

IDV

TPIT

DWG

TARA Visuals

TARA Setup

PowerGEM CA

Grid Map

Compare

Batch

Model QA

Results

Batch Inputs

Case Folder (.sav/.raw):

Browse

Script Folder (.idv/.py):

Browse

Output Folder:

Browse

Output Options

Output Suffix:

_scripted

☐ Include Subfolders

☐ Overwrite Outputs

Batch Run

Refresh Files

Run Batch

No batch run.

Case Outputs

Case	Status	Scripts	SAV Output	RAW Output	Error
------	--------	---------	------------	------------	-------

AELab25

Power System Engineering Workspace

v25.0

PSS/E Model Validation

☐ Dark mode

SSWG Model Validation

Run configurable QA checks against RAW, RAWX, or SAV cases and review the findings in the Results tab.

Case Scope

Case files		 Browse
Area numbers	33	Blank validates all areas.
Non-POI remote buses	70000-70999, 170000-170999	Numbers or ranges.

Run Controls

▶ Run Validation

■ View Results

⬇ Export Excel

No validation run.

Validation Checklist

Choose a profile or fine-tune individual checks by category.

14 of 17 checks selected

Defaults Select All Clear

Case Health

- ☒ Solvability and Base Violations
- ☒ PSS/E Powerflow Data Checks
- ☒ CNTB Messages
- ☐ TREE Islands

Topology and Bus Data

- ☒ Bus Type Inconsistencies
- ☒ Disconnected Buses
- ☒ 900k+ Buses
- ☒ Inconsistent Line End Voltages

Review ACCC, dynamics, model-validation, and TARA CA outputs in one workspace.

A25

AELab25

Power System Engineering Workspace

v25.0ResultsDark mode

ACCCDynamicsSSWVIDVTPITDWGTARA VisualsTARA SetupPowerGEM CAGrid MapCompareBatchModel QAResults

ACCC ResultsDynamic ResultsModel Validation ResultsTARA PowerGEM CA

Show Worst Only

Thermal ViolationsVoltage ViolationsVoltage Deviation ViolationsNot Converged

Filter: × Clear

Contingency	Branch	Rating (MVA)	Flow (MVA)	Loading (%)
-------------	--------	--------------	------------	-------------

Synthetic 1,499-row transfer example; no engineering conclusions are shown.

TARA Interactive Visualizer

-

Fit / 100%

+

TARA TRLIM

AC TRLim Compare

TARA ACCC

Batch_CA

Batch_Trlim Cross Compare

Line upgrade cost assumptions

138 kV (\$M / length unit)

2

345 kV (\$M / length unit)

4

Estimated cost = reported Length × voltage unit cost. Change either value to recalculate all tabs.

Load TARA Result

Sheet name

Top

20

Print PDF

Export Word

Export Excel

Loaded 1,499 rows from TARA_Example3.csv.

ROWS

1,499

FACILITIES

1,499

SELECTION

All buses (1,199 rows)

UPGRADE FLAGS

0

FIRST LIMIT

AC +108 MW

AC TRLIM RANGE

+108 to +1,895.2 MW

Worst AC TRLim per Receiving System

GEN_LOAD_AREA_040		+108	GEN_LOAD_AREA_040 30600 GEN_BUS_600 69 40300 GEN_BUS_300 69 3 GEN_CONT_200
GEN_LOAD_AREA_080		+108	GEN_LOAD_AREA_080 30500 GEN_BUS_500 69 40600 GEN_BUS_600 69 2 GEN_CONT_150
GEN_LOAD_AREA_120		+108	GEN_LOAD_AREA_120 30400 GEN_BUS_400 69 40200 GEN_BUS_200 69 1 GEN_CONT_100
GEN_LOAD_AREA_160		+108	GEN_LOAD_AREA_160 30300 GEN_BUS_300 69 40500 GEN_BUS_500 69 3 GEN_CONT_050
GEN_LOAD_AREA_200		+108	GEN_LOAD_AREA_200 30200 GEN_BUS_200 69 40100 GEN_BUS_100 69 2 Base Case
GEN_LOAD_AREA_240		+108	GEN_LOAD_AREA_240 30100 GEN_BUS_100 69 40400 GEN_BUS_400 69 1 GEN_CONT_200
GEN_LOAD_AREA_280		+108	GEN_LOAD_AREA_280 30000 GEN_BUS_000 69 40000 GEN_BUS_000 69 3 GEN_CONT_150
GEN_LOAD_AREA_010		+118.5	GEN_LOAD_AREA_010 30150 GEN_BUS_150 230 40250 GEN_BUS_250 230 3 GEN_CONT_050
GEN_LOAD_AREA_050		+118.5	GEN_LOAD_AREA_050 30050 GEN_BUS_050 230 40550 GEN_BUS_550 230 2 Base Case
GEN_LOAD_AREA_090		+118.5	GEN_LOAD_AREA_090 30650 GEN_BUS_650 230 40150 GEN_BUS_150 230 1 GEN_CONT_200
GEN_LOAD_AREA_130		+118.5	GEN_LOAD_AREA_130 30550 GEN_BUS_550 230 40450 GEN_BUS_450 230 3 GEN_CONT_150
			GEN_LOAD_AREA_170 30450

Worst AC TRLim per Case

Base Case		+108	30200 GEN_BUS_200 69 40100 GEN_BUS_100 69 2
GEN_CONT_050		+108	30300 GEN_BUS_300 69 40500 GEN_BUS_500 69 3
GEN_CONT_100		+108	30400 GEN_BUS_400 69 40200 GEN_BUS_200 69 1
GEN_CONT_150		+108	30500 GEN_BUS_500 69 40600 GEN_BUS_600 69 2
GEN_CONT_200		+108	30600 GEN_BUS_600 69 40300 GEN_BUS_300 69 3
GEN_CONT_001		+141.3	30003 GEN_BUS_003 138 40005 GEN_BUS_005 138 2
GEN_CONT_051		+141.3	30303 GEN_BUS_303 138 40505 GEN_BUS_505 138 1
GEN_CONT_101		+141.3	30403 GEN_BUS_403 138 40205 GEN_BUS_205 138 2
GEN_CONT_151		+141.3	30503 GEN_BUS_503 138 40605 GEN_BUS_605 138 3
GEN_CONT_201		+141.3	30603 GEN_BUS_603 138 40305 GEN_BUS_305 138 1
GEN_CONT_002		+161.4	30156 GEN_BUS_156 69 40260 GEN_BUS_260 69 3
GEN_CONT_052		+161.4	30256 GEN_BUS_256 69 40660 GEN_BUS_660 69 1
GEN_CONT_102		+161.4	30356 GEN_BUS_356 69 40360 GEN_BUS_360 69 2
GEN_CONT_152		+161.4	30456 GEN_BUS_456 69 40060 GEN_BUS_060 69 3
GEN_CONT_202		+161.4	30056 GEN_BUS_056 69 40560 GEN_BUS_560 69 2
GEN_CONT_003		+197.3	30159 GEN_BUS_159 138 40265 GEN_BUS_265 138 1

Synthetic Base, ALT1, and ALT2 comparison for interface demonstration only.

TARA Interactive Visualizer

-

Fit / 100%

+

TARA TRLIMAC TRLim CompareTARA ACCCBatch_CABatch_Trlim Cross Compare

Line upgrade cost assumptions

138 kV (\$M / length unit)2345 kV (\$M / length unit)4

Estimated cost = reported Length × voltage unit cost. Change either value to recalculate all tabs.

Load Batch CA Workbook

Detail rows250

Print PDF

Export Word

Export Excel

Loaded 3 Base/ALT scenarios from AELab25_DEMO_Base_ALT_CA.xlsx.

SCENARIOS3

BRANCH VIOLATIONS2

VOLTAGE VIOLATIONS1

TOTAL VIOLATIONS3

HIGHEST VIOLATING BRANCH LOADING120%

LOWEST VIOLATING VOLTAGE0.8800 p.u.

HIGHEST VIOLATING VOLTAGE0.8800 p.u.

Violations Across Base and ALT Scenarios

Branch violations

Voltage violations

Violations per ALT case

Base

ALT1

ALT2

1

2

0

Batch CA Scenario Comparison

Scenario	Branch Violations	Voltage Violations	Total Violations	Worst Branch Loading (%)	Worst Branch	Branch Contingency	Lowest Voltage (p.u.)	Lowest Voltage Bus	Lowest Voltage Contingency	Highest Voltage (p.u.)	Highest Voltage Bus	Highest Voltage Contingency
Base	1	0	1	104.6	LINE A	Base Case						
ALT1	1	1	2	120	LINE B	CONT 1	0.9	3,002	CONT 1	0.9	3,002	CONT 1
ALT2	0	0	0									

Branch Violations by Scenario